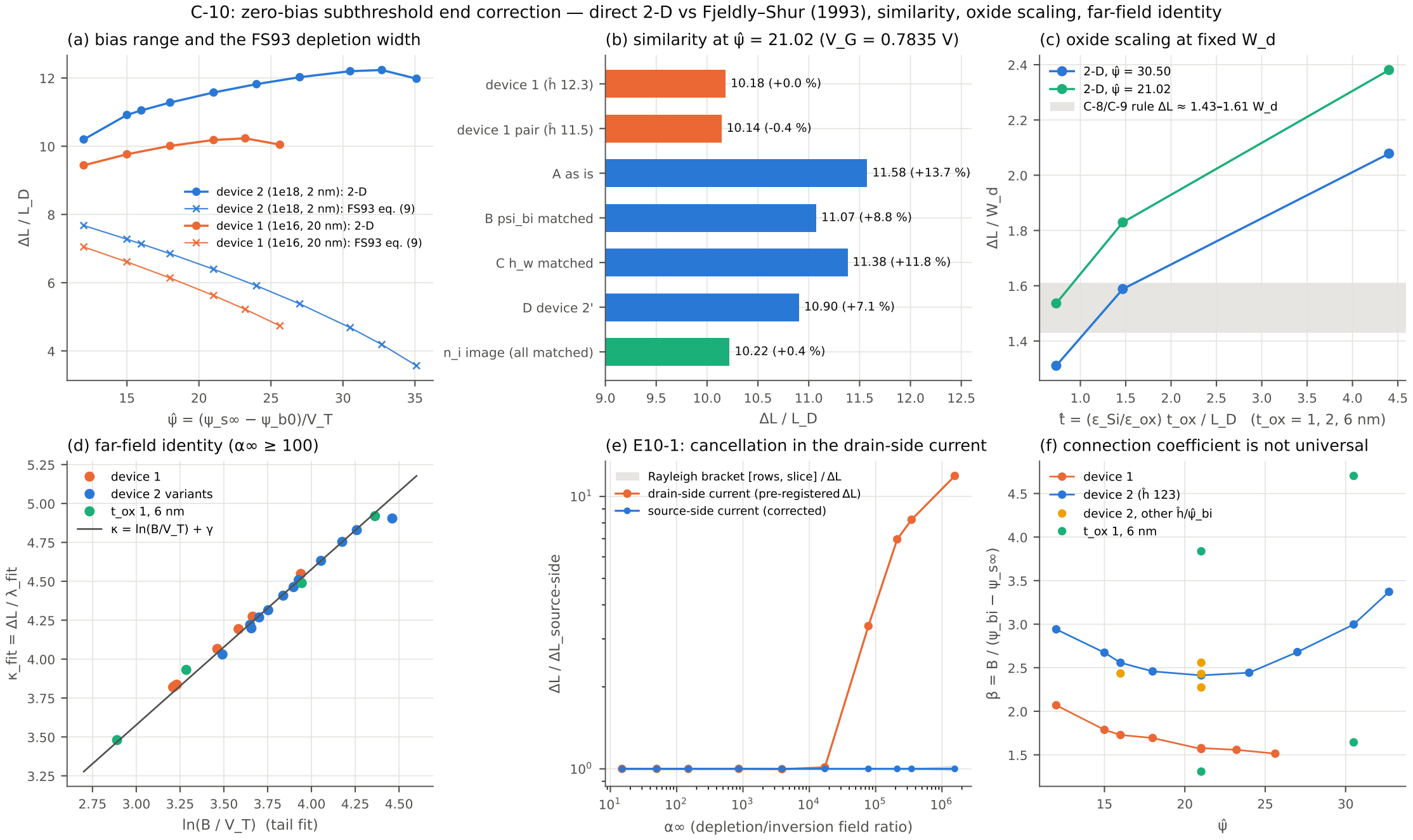
C-10: zero-bias subthreshold end correction — direct 2-D vs Fjeldly–Shur (1993), similarity, oxide scaling, far-field identity
(a) bias range and the FS93 depletion width
ΔL / L_D
ψ̂ = (ψ_s∞ − ψ_b0)/V_T
device 2 (1e18, 2 nm): 2-D
device 2 (1e18, 2 nm): FS93 eq. (9)
device 1 (1e16, 20 nm): 2-D
device 1 (1e16, 20 nm): FS93 eq. (9)
(b) similarity at ψ̂ = 21.02 (V_G = 0.7835 V)
device 1 (ĥ 12.3)
10.18 (+0.0 %)
device 1 pair (ĥ 11.5)
10.14 (-0.4 %)
A as is
11.58 (+13.7 %)
B psi_bi matched
11.07 (+8.8 %)
C h_w matched
11.38 (+11.8 %)
D device 2'
10.90 (+7.1 %)
n_i image (all matched)
10.22 (+0.4 %)
ΔL / L_D
(c) oxide scaling at fixed W_d
2-D, ψ̂ = 30.50
2-D, ψ̂ = 21.02
C-8/C-9 rule ΔL ≈ 1.43–1.61 W_d
ΔL / W_d
t̂ = (ε_Si/ε_ox) t_ox / L_D (t_ox = 1, 2, 6 nm)
(d) far-field identity (α∞ ≥ 100)
device 1
device 2 variants
t_ox 1, 6 nm
κ = ln(B/V_T) + γ
κ_fit = ΔL / λ_fit
ln(B / V_T) (tail fit)
(e) E10-1: cancellation in the drain-side current
Rayleigh bracket [rows, slice] / ΔL
drain-side current (pre-registered ΔL)
source-side current (corrected)
ΔL / ΔL_source-side
α∞ (depletion/inversion field ratio)
(f) connection coefficient is not universal
device 1
device 2 (ĥ 123)
device 2, other ĥ/ψ̂_bi
t_ox 1, 6 nm
β = B / (ψ_bi − ψ_s∞)
ψ̂